Parametrizações Físicas na Previsão Numérica de Tempo e Clima

Paulo Yoshio Kubota

3.2 Equações contínuas em coordenadas esféricas

As equações para as variáveis acopladas são (Eqs. 6 a 10 do artigo):

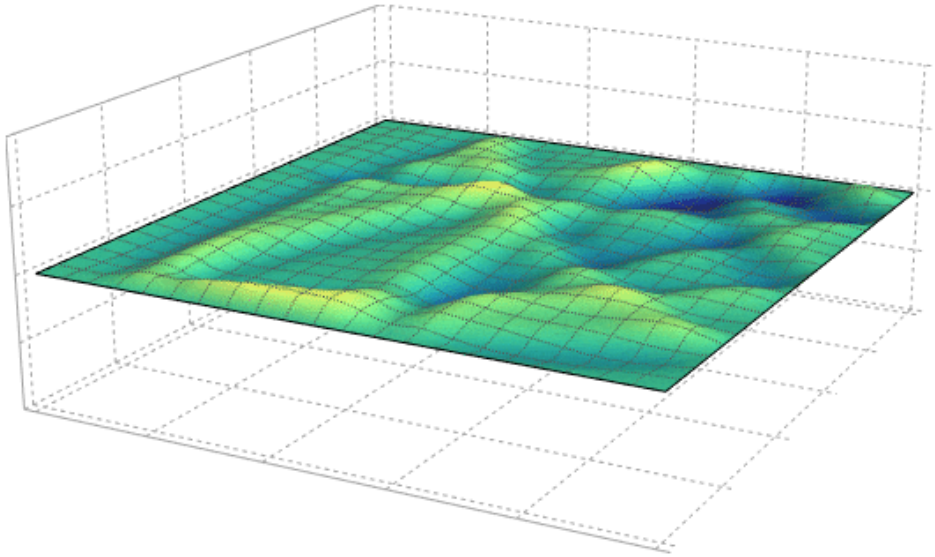
$$\frac{\partial \tilde{\rho}_d \vec{V}_H}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{P}{\zeta_z}\right) - \frac{\partial(z_H P)}{\partial \zeta}\right] - \eta\hat{\boldsymbol{k}} \times \tilde{\rho}_d \vec{V}_H - \vec{V}_H \nabla_\zeta \cdot \vec{V}_H - \left[\frac{\partial(\Omega \vec{V}_H)}{\partial \zeta}\right] - \tilde{\rho}_d \nabla_\zeta K - \tilde{\rho}_d \beta_d \left(ew\hat{\imath} + \frac{\vec{V}_H \omega}{r}\right) + \tilde{\rho}_d \frac{F}{\rho_d}$$

$$\frac{\partial W}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\frac{\partial P}{\partial \zeta} + g\left(\frac{r_0}{r}\right)^2 \tilde{\rho}_m\right] - \left(\nabla \cdot \vec{V} w\right)_\zeta + \tilde{\rho}_d eu + \tilde{\rho}_d \beta_d \left(\frac{\vec{V}_H \cdot \vec{V}_H}{r} + e\vec{V}_H \cdot \hat{\imath} +\right) + \frac{\tilde{\rho}_d F_w}{\rho_d}$$

$$\frac{\partial \Theta}{\partial t} + \left(\nabla \cdot \vec{V}\theta_m\right)_\zeta = F_\Theta \qquad (3.6)$$

$$\frac{\partial Q_j}{\partial t} + \left(\nabla \cdot \left(\vec{V} q_j\right)\right)_\zeta = F_{Q_j} \qquad (3.8)$$

$$\frac{\partial \tilde{\rho}_d}{\partial t} = -(\nabla \cdot V)_\zeta \qquad (3.7)$$



$$p = p_0\left(\frac{R_d \Theta_m}{p_0 z_\zeta}\right)^\gamma.$$

Diabatic Forcing (perturbations force)

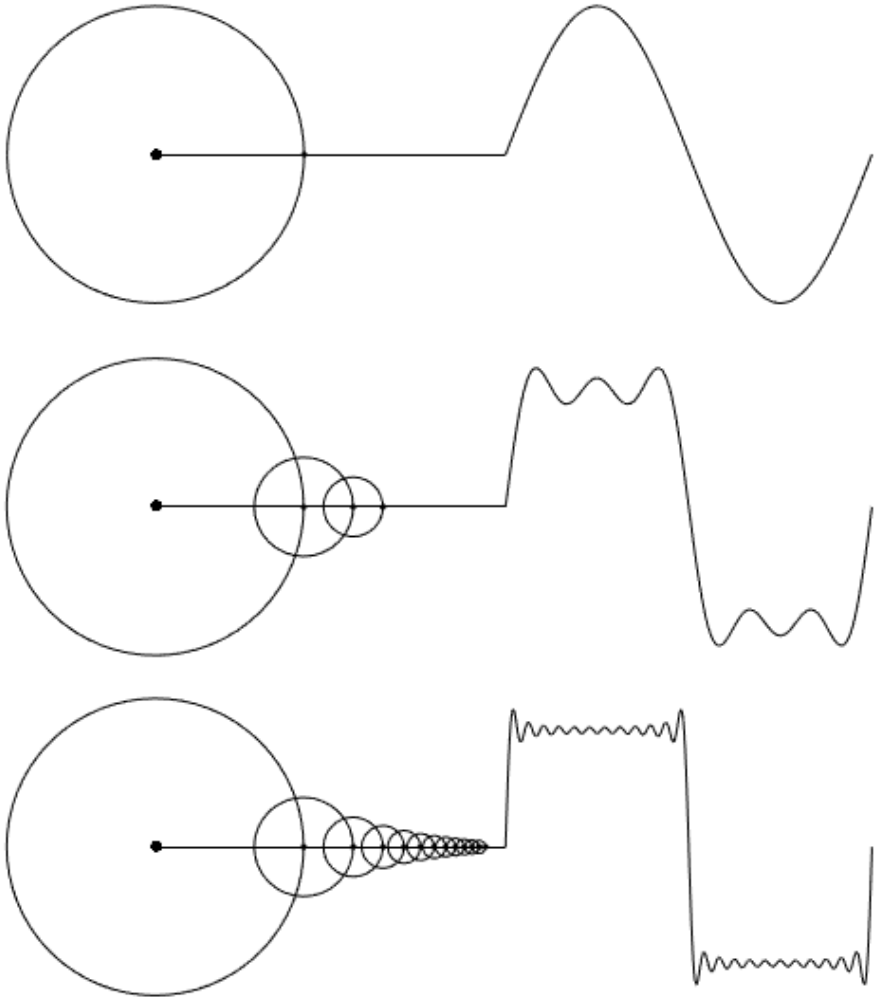
As equações para as variáveis acopladas são (Eqs. 6 a 10 do artigo):

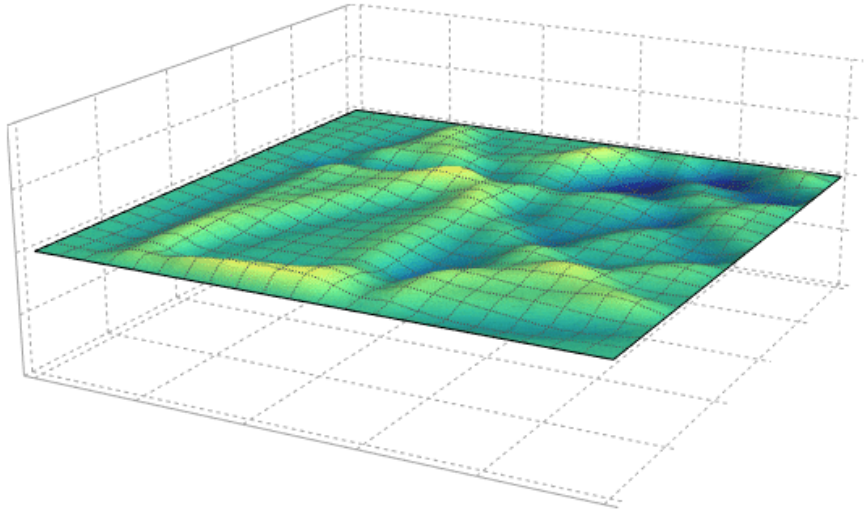
F

F_w

F_Θ

F_{Q_j}





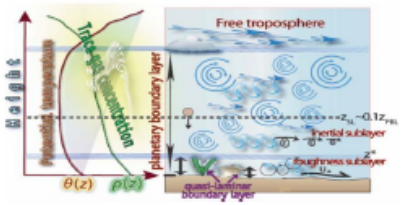
Surface Forcing

PARAMETRIZAÇÕES

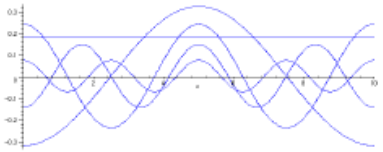
Superfície

$$|\tau_z| = \rho\sqrt{[(\overline{w'u'})^2 + (\overline{w'v'})^2]}$$

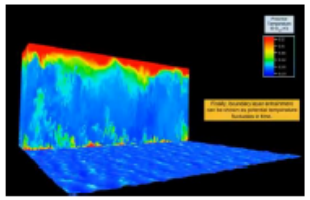
J. Sommar et al.



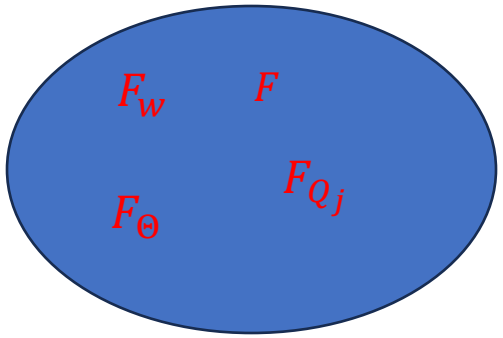
$$H = \rho_a c_{pd} \overline{w'\theta_v}'$$

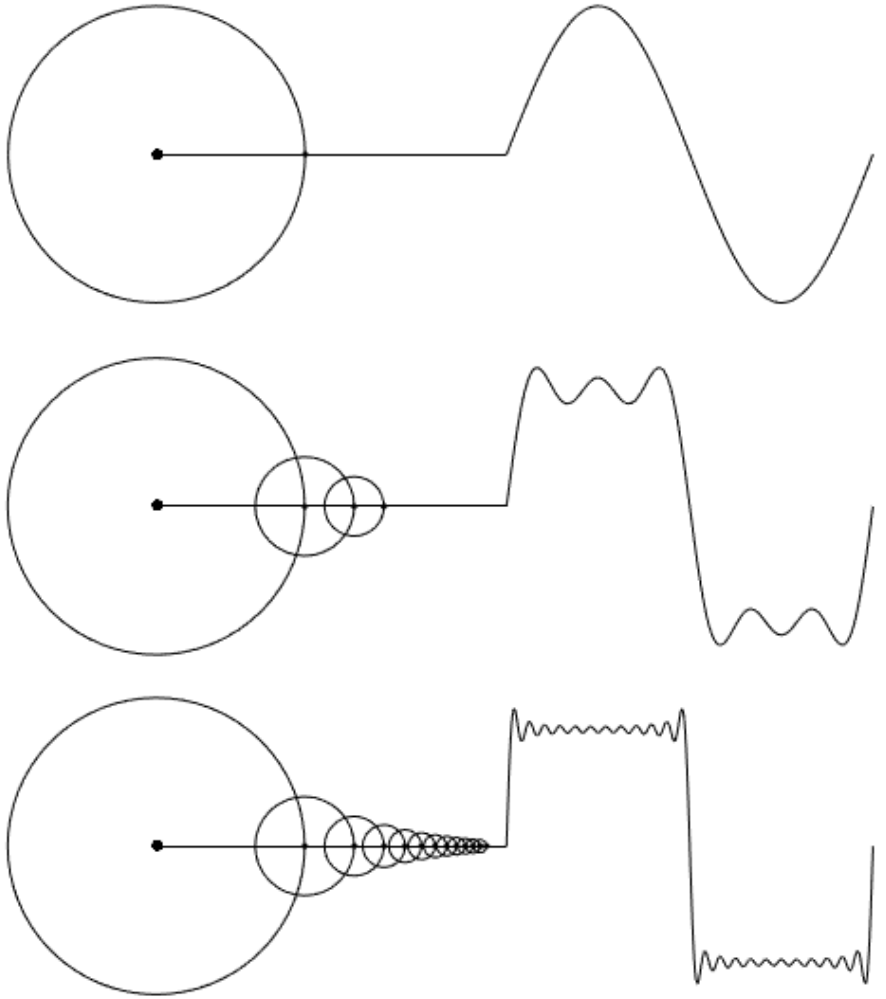


$$\mathrm{Ef} = \rho_a L_v \overline{w'q'}$$



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Convection Forcing

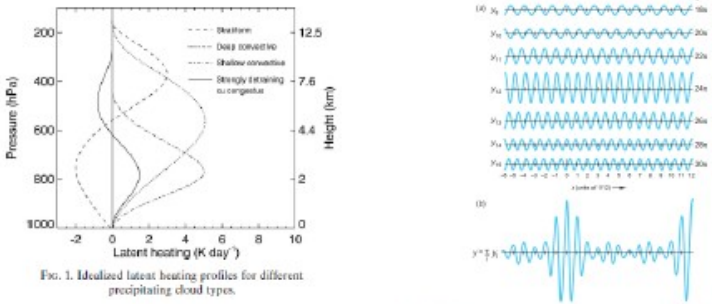
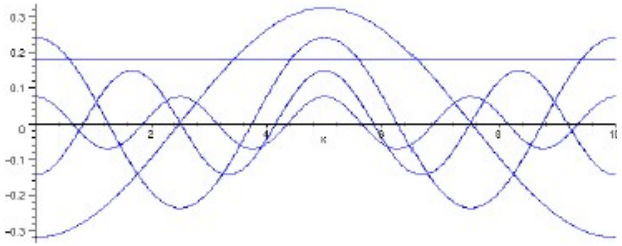
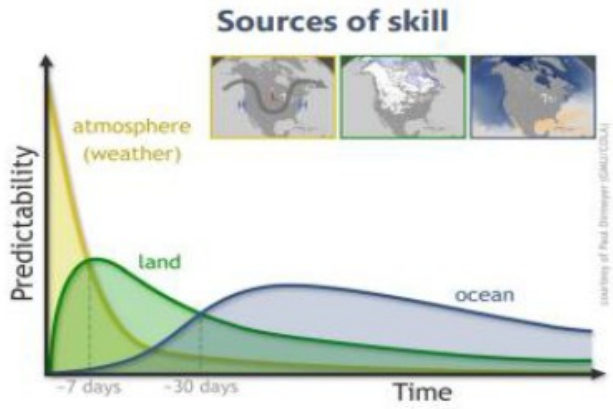
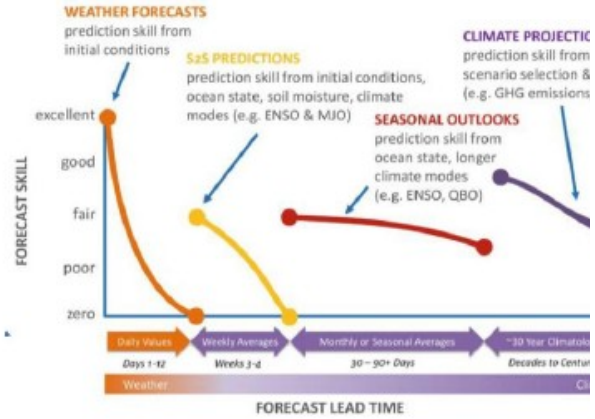
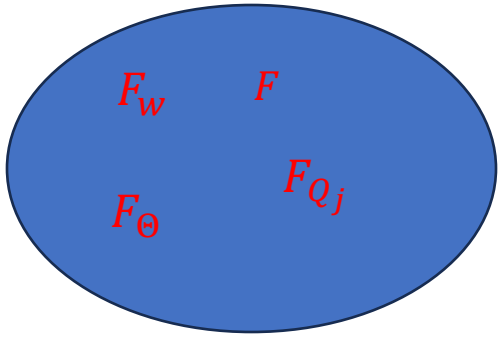


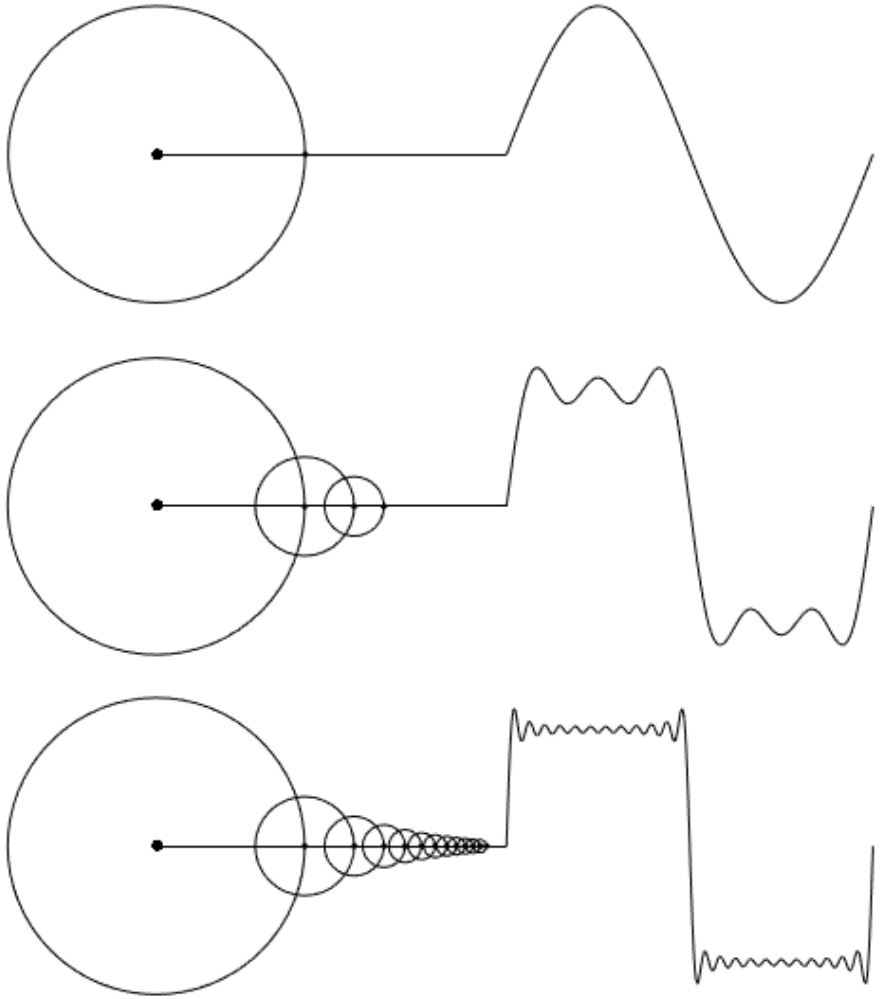
Fig. 1. Idealized latent heating profiles for different precipitating cloud types.





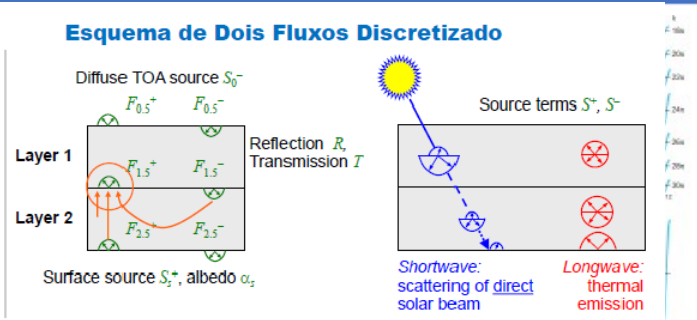


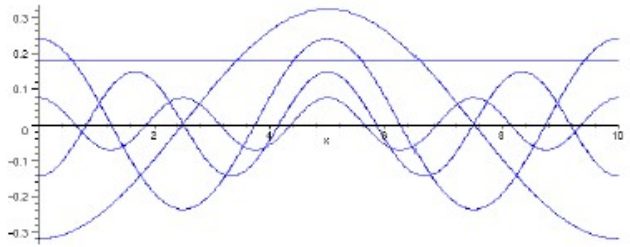


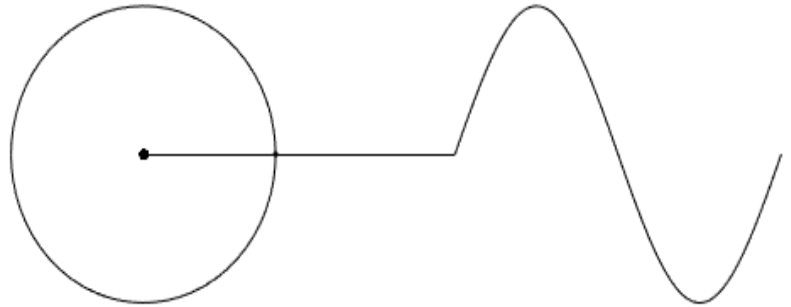


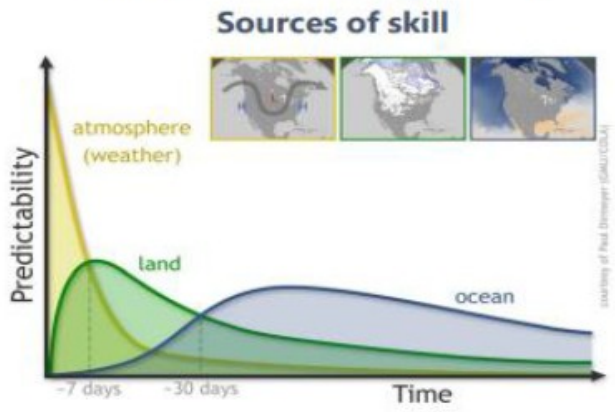
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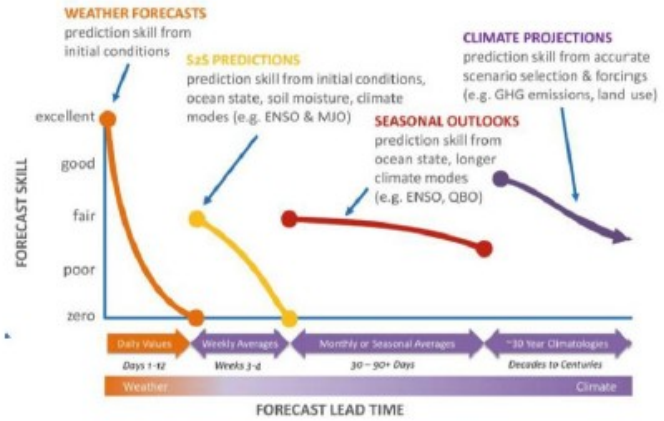
Radiative Forcing

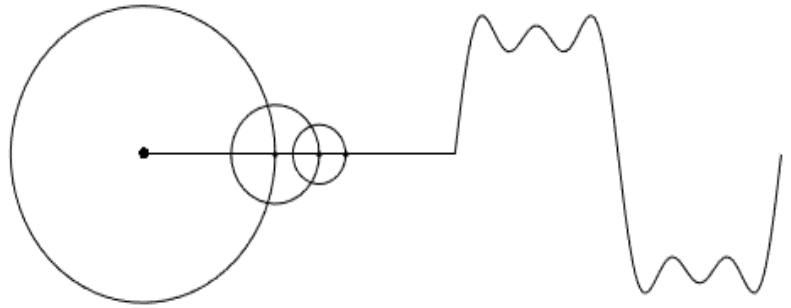


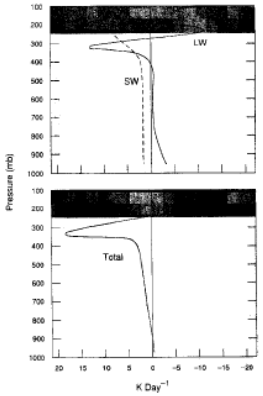












$$F_{i+0.5}^{+} = T_i F_{i+1.5}^{+} + R_i F_{i-0.5}^{-} + S_i^{+}$$

$$F_{i+1.5}^{-} = T F_{i+0.5}^{-} + R_i F_{i+1.5}^{+} + S_i^{+}$$

$$\frac{\partial(\overline{T})}{\partial t} = -\frac{1}{\overline{\rho} c_p} \frac{\partial \overline{F_j}}{\partial x_j}$$

$$\frac{\partial T}{\partial t} = \frac{1}{\rho C_p} \frac{\partial L^*}{\partial z} = \frac{1}{1 * 1004} \frac{-300 - -100}{15000} \sim -1.15\, K/day$$

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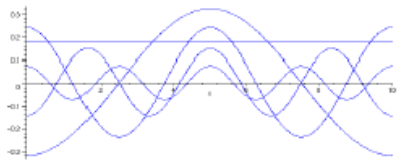
Turbulence Forcing

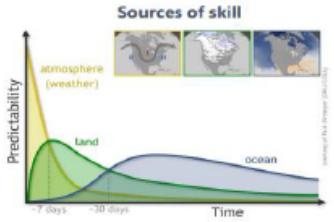
O problema da parametrização de camada limite planetária é determinar como será feita a difusão a cada instante do tempo

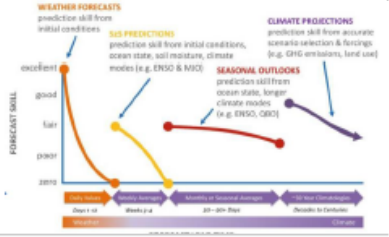
K(z, estabilidade, nebulosidade, radiação, turbulencia, etc)

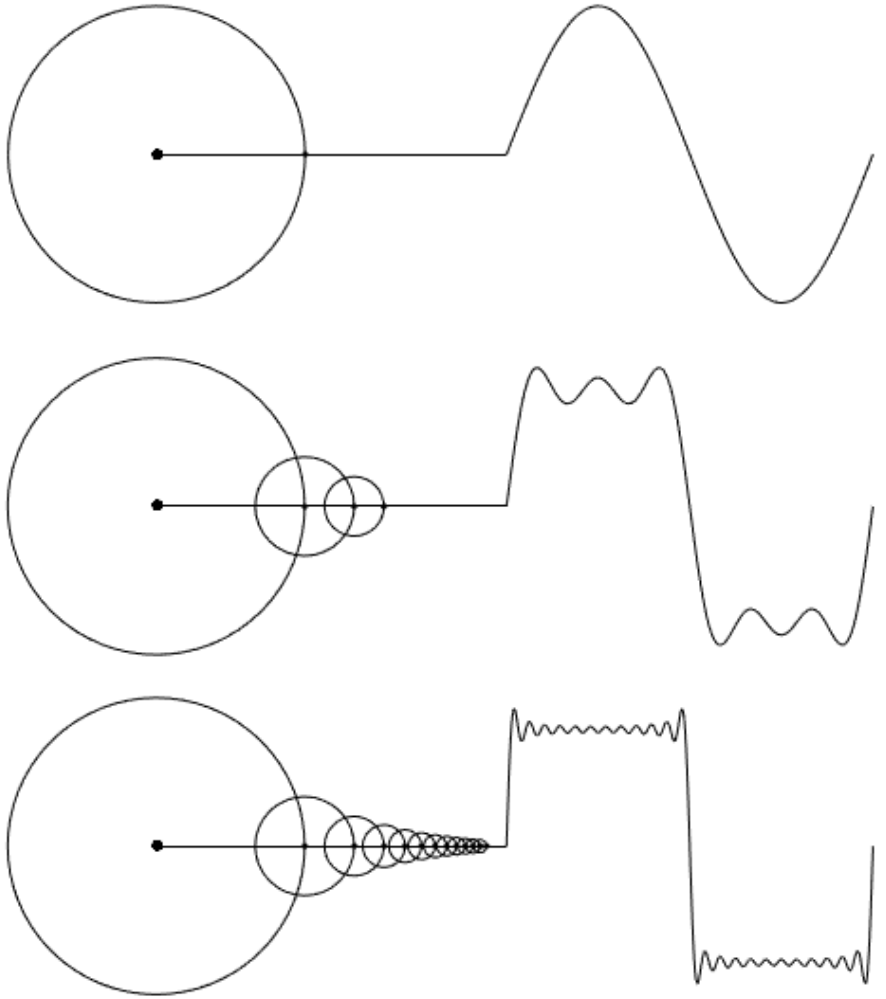
K=K(z, estabilidade, nebulosidade, radiação, turbulencia, etc)??????
Precisa saber todos os fenômenos físicos que afetam o perfil vertical de de K_M, K_H

Como parameterizar os momentum de 2 ordem $w'\varphi'$, K_M, K_H









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Equações contínuas em coordenadas esféricas

$$\frac{\partial \tilde{\rho}_d \vec{V}_H}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\nabla_\zeta\left(\frac{P}{\zeta_z}\right) - \frac{\partial(z_H P)}{\partial \zeta}\right] - \eta\hat{k}\times\tilde{\rho}_d\vec{V}_H - \vec{V}_H\nabla\zeta\cdot\vec{V}_H - \left[\frac{\partial(\Omega\vec{V}_H)}{\partial\zeta}\right] - \tilde{\rho}_d\nabla_\zeta K - \tilde{\rho}_d\beta_d\left(ew\hat{\imath} + \frac{\vec{V}_H\omega}{r}\right) + \tilde{\rho}_d\frac{F}{\rho_d}$$

$$\frac{\partial W}{\partial t} = -\frac{\rho_d}{\rho_m}\left[\frac{\partial P}{\partial\zeta} + g\left(\frac{r_0}{r}\right)^2\tilde{\rho}_m\right] - (\nabla\cdot\vec{V}w)_\zeta + \tilde{\rho}_d eu + \tilde{\rho}_d\beta_d\left(\frac{\vec{V}_H\cdot\vec{V}_H}{r} + e\vec{V}_H\cdot\hat{\imath} + \right) + \frac{\tilde{\rho}_d F_w}{\rho_d}$$

$$\frac{\partial\Theta}{\partial t} + \left(\nabla\cdot\vec{V}\theta_m\right)_\zeta = F_\Theta \qquad (3.6)$$

$$\frac{\partial Q_j}{\partial t} + \left(\nabla\cdot(\vec{V}q_j)\right)_\zeta = F_{Q_j} \qquad (3.8)$$

$$\frac{\partial\tilde{\rho}_d}{\partial t} = -(\nabla\cdot V)_\zeta \qquad (3.7)$$